

FOSS

in Academic Institutions



In recent years, we have seen open source being widely adapted by many organisations around the world—not only because of its cost-effectiveness, but also because of its stability, reliability, flexibility and the freedom to modify it. Every sector (the government, the private sector or the scientific institutions) uses it with great success. One sector where open source has evolved a lot in recent years is academia. These days, many academic institutes' IT infrastructures are implemented using FOSS-based solutions.

This article aims to engage systems analysts of universities, colleges and schools, as well as student volunteers who want to implement some infrastructural services at their institutes. Here, we will cover how FOSS-based solutions can help them implement some specific facilities in their institutions that are not only cost-effective, but can be used to increase productivity and improve utilisation of the available resources.

Off-campus resource sharing

These days, a lot of institutions subscribe to online journals. These subscriptions are costly, ranging from several thousands to several lakhs of rupees per publisher, per year. These subscriptions are generally based on IP address-based authentication. These are the allocated IP addresses of the institutions. If the campus is non-residential, most of these resources are left underutilised, as they cannot be accessed from any network except the institution's own. During office hours, users (faculty and students) don't have much time to go through these resources, because of daily scheduled work. Now, in order to make these resources available to users after working hours from their homes, we can deploy some open source solutions:

- **Squid proxy with OpenLDAP:** A Squid proxy will allow students to use the institution's proxy from outside the campus, after being authenticated against the OpenLDAP server. Once the users are authenticated, they will be routed to the publisher's website from the institution's server (which is authenticated by the publisher), allowing them to browse research articles based on IP address authentication.
- **OpenVPN with OpenLDAP:** OpenVPN will create a secure connection to the server after the users have been authenticated against the OpenLDAP server, and after that, they can browse the journals.

Authentication services

These days, a lot of services like integrated library management systems, research repositories, learning management systems, off-campus resource sharing and other daily-use office applications are used by academic institutions in their day-to-day operations. It is very difficult for users to remember the credentials for all these services separately. To solve this problem, one can set up an authentication server that will be used by different services in order to authenticate users:

- **OpenLDAP:** This can be used as a back-end server to authenticate users for all institutional infrastructure services.
- Other alternatives like Apache directory server, OpenDS and OpenDJ also exist.
- Identity providers like Shibboleth can also be used with any LDAP server as the back-end.

Since the majority of FOSS projects already have built-in support for authenticating users against LDAP, this authentication service is preferred over others.

Integrated library management software

Every university these days has resource centres containing thousands of books, digital assets and old document archives. Also, as current library budgets range from several lakhs to several crore rupees, there's a need for a solution that can help in financial audits and budget-related issues.

- **Koha:** Koha is a good integrated library management software (ILS) that is used across the globe. It provides facilities like tagging and RSS feeds, the union catalogue facility, customisable searches, circulation and borrower management. It also offers a full acquisitions system with budgets and pricing information (supplier and currency conversion), apart from the ability to cope with